Section 4. Pre-registered analysis (AsPredicted #136723)

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Generative Artificial Intelligence and Creative Production (#136723)

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1) Have any data been collected for this study already?
No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?
How does the availability of generative AI to assist with a creative task (i.e. writing a short story) affect the (self-)evaluation of the creative output by creators and third-party evaluators?

3) Describe the key dependent variable(s) specifying how they will be measured.
We are interested in assessing the following dependent variables, which will be measured for both creators and third-party evaluators:
- Novelty: an index of three questions on a scale of 1 to 9
- Usefulness: an index of three questions on a scale of 1 to 9

We will create aggregate indices for these measures, as well as look at their individual components.

4) How many and which conditions will participants be assigned to?
There are three conditions in the study:
- "Human only" condition where the project creator does not get any AI assistance
- "Hybrid" condition where the creator has the opportunity to access one short prompt for a story idea from OpenAI's ChatGPT API
- "Hybrid+" condition where the creator has the opportunity to access up to five short prompts for a story idea from OpenAI's ChatGPT API

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
We will run OLS regressions predicting novelty and usefulness by condition, and run these regressions for both creators and evaluators.

We will run robustness tests for each of these, which will include different econometric specifications and variants (e.g., sub-items, discretization) of the outcome measures.

Note: While the Hybrid and Hybrid+ conditions technically differ in their capabilities (the latter allows for more AI prompts), we plan to combine the two conditions into one joint condition for our main analysis if the main outcome variables in those two conditions are not statistically significant from each other. (We will still report the existence of all three conditions and a results breakdown by all three conditions in the appendix.)

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
We will exclude all participants that did not finish the study for analysis purposes. We will also drop respondents in the "Human only" condition that acknowledged that they used generative AI to assist with their responses.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.
For each condition, we will collect n=100 creators per condition who complete the study, for a total of n=300 creators across the three conditions. Then, we will collect n=600 third-party evaluators (each of which evaluates six stories drawn at random from the creator conditions).

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)
We have additional exploratory outcome variables about each story (e.g. enjoyment of the story, how well written, funny or boring the story is, etc.) on a scale from 1 to 9, for which we will study treatment effects similar to our main analysis.

We have collected a number of variables that we will use to look at heterogeneous effects including the creativity of the respondent (through a creativity task prior to the writing task and through self-ratings), their prior experience with generative AI technologies, and demographic information (e.g. gender, education, employment and income).

We will also consider non-linear relationships, by discretizing the outcomes and running linear probability models.

Section 5. Writer study screenshots